

TANVIR AHMED

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Education

Bachelor of Science in Computer Science and Engineering **2022 – Dec 2026 (expected)**

North South University, Dhaka, Bangladesh

Department of Electrical and Computer Engineering

CGPA: **3.02** / **4.00**

Higher Secondary Certificate (HSC), Science

January 2021

Milestone College, Dhaka

GPA: **5.00** / **5.00**

Research Interests

- Efficient Multimodal Inference
- Computer Vision and Video Understanding
- Calibration, Abstention and Uncertainty
- Quantum Machine Learning
- Explainable AI (XAI)
- Edge AI
- Agentic Systems
- Full-Stack System Architecture

Publications

Closed-Book Knowledge Base Construction with Calibrated Abstention and Numeric Self-Consistency

Accepted

Tanvir Ahmed. *LM-KBC 2026 shared task (5th edition), Automated Knowledge Base Construction (AKBC) at EMNLP 2026.*

- Set-valued knowledge base construction with an open-weight model of at most 32B parameters, no retrieval and no fine-tuning; treats deciding *when to answer* as a first-class decision and beats the official baseline on five of the six relations.
- Code: github.com/minhaz-42/lm-kbc-2026

SoundSentry: A Low-Cost Edge-AI Acoustic Event Detection and Direction-Finding System on STM32

Under review

Tanvir Ahmed, Sanzana Afrin Tonny, Husain Kabir Sifat. *International Conference on Electrical and Computer Engineering (ICECE) 2026, Dhaka.*

- STM32F103 prototype with a three-microphone array, ADC-DMA acquisition, on-device classification over 11 acoustic features, servo-based direction response and SD-card logging.
- Code: github.com/minhaz-42/SoundSentry-Acoustic-Surveillance-Direction-Finding-System

Research Projects

Frame-Aware Multi-Model Scheduling for Efficient Video Captioning

Senior Design Project, 2026

Routes video captioning between a 4B and a 27B vision-language model so the large model is spent only where the footage is hard. Lead author of four.

- Frames are stitched into mosaics, encoded by a frozen CLIP ViT-B/32 and scored for visual complexity by a small graph attention network trained without human labels.
- Answers a 19-clip benchmark in fewer requests than either model alone while scoring above the small model on every judged axis.

Technologies: Python, PyTorch, PyTorch Geometric, CLIP, Django, Gemma 3, Qwen3-VL

Quantum-Enhanced Driver Distraction Detection

Directed Research, 2026

A hybrid classical-quantum pipeline with uncertainty-aware risk scoring, tested under a protocol where only the classification head differs.

- The variational quantum head reaches accuracy parity with its matched classical control and cuts expected calibration error by 37% relative.
- The report states what the evidence does not support, including that the calibration gain cannot be attributed to the circuit.

Technologies: Python, PyTorch, PennyLane, EfficientNet, scikit-learn

Bangla Mental Health Conversational Agent

Ongoing

AI-powered mental health chatbot for Bangla-speaking users providing context-aware, empathetic conversations.

- Fine-tuned models: Gemma-3-Bangla, Qwen-2.5-0.5B-Bangla
- Features: voice I/O, mood tracking, conversation history, multi-language UI
- Technical: Django backend, OAuth authentication, Apple MPS GPU acceleration

Technologies: Python, Django, PyTorch, Transformers, REST APIs, SQLite

Observing Is Not Free: the decision value of visual observations in vision-language agents

In preparation

Measures whether an agent's observation-selection criterion ranks observations the way their realised effect on the answer does, using a counterfactual table of answers before and after each candidate observation.

Technologies: Python, Vision-Language Models

Knowing What You No Longer Know: calibrated belief validity for persistent world memory

In preparation

Separates whether a belief is still true from how confident the model was when it formed it, and builds a benchmark of elapsed-time-matched pairs to score it.

Technologies: Python, PyTorch, HD-EPIC, AI2-THOR, ProcTHOR

SimCert: A Classical-Simulability Audit of Trained QNLP Text Models

Abstract submitted, QTML 2026

Re-simulates trained quantum-NLP circuits under matrix-product-state truncation to test whether the quantum component is load-bearing. Code: github.com/minhaz-42/simcert-qnlp

Technologies: Python, PennyLane, lambeq, OpenQASM, Hydra

Sequential Video Belief

In progress

Whether a video-language model's own stated answer impairs its ability to revise that answer when later visual evidence contradicts it.

Technologies: Python, Vision-Language Models

Uncertainty-Guided Language-Grounded 3D Scene Graphs from Sparse Views **In progress**

Frozen geometry and vision-language models with a lightweight graph reasoning module, for 3D object segmentation and spatial relations from a few casual images.

Technologies: Python, PyTorch, VGGT, SAM 2, CLIP, DINOv2

Technical Projects

Pakna Council – AI Council Platform 2026

Local-first web app running multiple LLMs in parallel on one prompt through Ollama, with model comparison and vision workflows.

Links: [Source](#) · [Live site](#)

Technologies: Python, Django, React, Docker, Ollama, WebSocket

SHAND – Explainable Assumption Detector 2026

Rule-based system for analysing assumptions in English text; includes visual analytics and export.

Links: [Source](#) · [Live site](#)

Technologies: Python, NLP, HTML, JavaScript, D3.js

ISKA – Privacy-First Document Analysis Assistant 2026

Document analysis assistant with a Chrome extension for use while reading.

Links: [Source](#) · [Live site](#)

Technologies: TypeScript, Chrome extension APIs

Nestledown – 3D Exploration Game 2026

Cozy 3D exploration game built solo in Godot 4 with every asset script-generated; a parallel build runs on a custom C++20 engine with a Metal renderer.

Links: [Announcement trailer](#)

Technologies: Godot 4, Blender, C++20, Metal, SDL3, CMake

Entangled Fusion Quantum Circuits for Vision–Language Alignment 2026

Experimental multimodal platform combining vision analysis with quantum-inspired fusion and natural language reasoning.

Links: [Source](#)

Technologies: Python, Django, PennyLane, Ollama, OpenCV, REST APIs

SUMBA – 3D Sign Language Translation 2026

Real-time sign-to-text translation using computer vision with mobile integration and offline functionality.

Links: [Source](#)

Technologies: Django, MediaPipe, Transformers, React Native

FitWell – AI Nutrition Assistant 2025

AI-powered meal planning, BMI tracking and health insights.

Links: Source

Technologies: Django, Python, Qwen, SQLite, Chart.js

DG Healthcare Management System 2024

Hospital management platform with appointments, billing and role-based access.

Links: Source

Technologies: HTML5, CSS3, PHP, MySQL, JavaScript

CSE215 Java Application 2024

OOP-focused Java application with modular design, MVC pattern and unit testing.

Links: Source

Technologies: Java, JavaFX, MySQL, JUnit

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, PHP, C/C++, SQL, HTML, CSS, L^AT_EX

Machine Learning & AI: Deep Learning, Computer Vision, NLP, LLMs, Transformers, Vision-Language Models, MediaPipe, Quantum ML, Fine-tuning, Calibration

Frameworks & Libraries: PyTorch, PyTorch Geometric, TensorFlow, scikit-learn, Django, React, Node.js, JavaFX, PennyLane, lambeq, OpenCV, CLIP, Hydra

Databases: MySQL, SQLite, PostgreSQL, MongoDB

Software Engineering: OOP, Design Patterns, Agile/Scrum, REST APIs, WebSocket, Docker, CI/CD, Unit Testing

Embedded & Graphics: STM32, Embedded C, ADC-DMA acquisition, Godot 4, Blender, Metal, SDL3, CMake

Tools & Platforms: Git, VS Code, Jupyter, L^AT_EX, Linux, Ollama, Apple MPS

Specializations: Full-Stack Development, System Architecture, LLM Orchestration, Edge AI Deployment

Extracurricular Activities

IEEE NSU Student Branch, General Member 2023 – Present

- Participated in technical workshops, seminars, and community outreach programs

NSU Trimodal AI Lab, Research Member 2025 – Present

- Engaged in research discussions and collaborative experimentation on multimodal and agentic AI systems
- Contributed to idea validation, prototyping, and literature exploration in emerging AI domains

References

Dr. Md. Adnan Arefeen

Assistant Professor, Dept. of ECE, North South University

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